

50/60 kW, 1000 Vdc String Inverters for North America

The CPS 50/60 kW three-phase string inverters are designed for ground mount, rooftop and carport applications. The units are high performance, advanced, and reliable inverters designed specifically for the North American environment and grid. High efficiency at 98.8% peak and 98.5% CEC, wide operating voltages, broad temperature ranges, and a NEMA Type 4X enclosure enable this inverter platform to operate at high performance across many applications.

The CPS 50/60KTL products ship with either the Standard wire box or the Rapid Shutdown wire box, each fully integrated and separable with touch-safe fusing, monitoring, and AC and DC disconnect switches. The integrated PLC transmitter in the Rapid Shutdown wire box enables PVRSS certified module-level rapid shutdown when used with APS RSD-S-PLC/RSD-D products.



CPS SCA50KTL-DO/US-480
CPS SCA60KTL-DO/US-480

Key Features

- PVRSS certified for rapid shutdown
- 55 & 66 kVA rating allows max rated active power @ ± 0.91 PF
- Selectable max. AC apparent power of 50/55 kVA and 60/66 kVA
- UL-1699B compliant arc-fault circuit protection
- 15-90° mounting orientation for low profile roof installs
- Integrated AC and DC disconnect switches
- 3 MPPTs with 5 inputs each for maximum flexibility
- NEMA Type 4X outdoor rated enclosure
- UL 1741-SA certified to CA Rule 21, including SA8 - SA18
- UL 1741-SB and IEEE 1547-2018 certified
- Separable wire-box design for fast service
- Standard 10-year warranty with extensions up to 20 years



50/60KTL Standard Wire Box



50/60KTL Rapid Shutdown Wire Box

Model name	CPS SCA50KTL-DO/US-480	CPS SCA60KTL-DO/US-480
DC Input		
Max. PV power	90 kW (33 kW per MPPT)	
Max. DC input voltage	1000 Vdc	
Operating DC input voltage range	200-950 Vdc	
Startup DC input voltage / power	330 V / 80 W	
Number of MPPTs	3	
MPPT voltage range for Pnom @ PF>0.99	480-850 Vdc	540-850 Vdc
Max. operating current per MPPT	36 A per MPPT	38 A per MPPT
Production limits	20.7 kW per MPPT or 50 kW total output	22.2 kW per MPPT or 60 kW total output
Max. PV short circuit current ¹	163.2 A (54.4 A per MPPT)	
Number of DC inputs	15 inputs, 5 per MPPT	
DC disconnection type	Load-rated DC switch	
DC surge protection	Type II MOV	
AC Output		
Rated AC output power @ PF>0.99 to ±0.91 ²	50 kW	60 kW
Max. AC apparent power (selectable ³)	50 / 55 kVA	60 / 66 kVA
Rated output voltage	480 Vac	
Output voltage range ⁴	422-528 Vac	
Grid connection type	3Φ / PE / N (neutral optional)	
Max. AC output current @ 480 Vac	60.2 A (@ 50 kVA) / 66.2 A (@ 55 kVA)	72.2 A (@ 60 kVA) / 79.4 A (@ 66 kVA)
Rated output frequency	60 Hz	
Output frequency range ⁴	57-63 Hz	
Power factor	>0.99 (±0.8 adjustable)	
Current TRD @ rated load	< 3%	
Max. fault current contribution (1 cycle RMS)	64.1 A (1.06/0.88 PU)	
Max. OCPD rating	125 A	125 A
AC disconnection type	Load-break rated AC switch	
AC surge protection	Type II MOV	
System and Performance		
Topology	Transformerless	
Max. efficiency	98.8%	
CEC efficiency	98.5%	
Night consumption	< 1 W	
Environment		
Enclosure protection degree	NEMA 4X	
Cooling method	Variable speed cooling fans	
Operating temperature range ⁵	-22°F to 140°F (-30°C to 60°C)	
Non-operating temperature range	-40°F to 158°F (-40°C to 70°C)	
Operating humidity	0-100%	
Operating altitude	13123 ft / 4000 m (derating from 9843 ft / 3000 m)	
Audible noise	<60 dBA @ 1 m and 77°F (25°C)	
Display and Communication		
User interface and display	LCD+LED	
Inverter communication	Modbus RS485	
Modbus data mapping	CPS	
Mechanical		
Dimensions (H × W × D)	39.4 × 23.6 × 10.24 in (1000 × 600 × 260 mm)	
Weight	Inverter: 123.5 lb (56 kg) Wire box: 33 lb (15 kg)	
Mounting / installation angle ⁶	15 to 90 degrees from horizontal (vertical or angled)	
AC termination	M8 stud type terminal block (wire range: #6-3/0 AWG CU/AL; lugs not supplied)	
DC termination ⁷	Screw clamp, negative busbar (RSD version ⁷), wire range: #14-#6 AWG CU	
Fused string inputs (5 per MPPT)	RSD ⁷ and Standard wire box: 25 A fuses provided (fuse values up to 30 A acceptable)	
Safety		
Certifications and standards	UL 1741-SA/SB Ed. 3, UL 1699B, UL 1998, CSA-C22.2 NO.107.1-01, IEEE 1547-2018, FCC Part 15	
Selectable grid standards	IEEE 1547a-2014, IEEE 1547-2018 ⁸ , CA Rule 21, ISO-NE, HECO	
Smart-grid features	Volt-RideThru, Freq-RideThru, Ramp-Rate, Specified-PF, Volt-VAR, Freq-Watt, Volt-Watt	
Warranty		
Standard	10 years	
Extended terms	15 and 20 years	

1) The sum of parallel-connected PV module short-circuit currents.

2) Active power derating begins at PF = ± 0.91 to ± 0.80 when max AC apparent power is set to 55 or 66 kVA.

3) Inverters are factory set to 50 kVA and 60 kVA by default. Contact CPS to enable the higher kVA setting.

4) The "output voltage range" and "output frequency range" may differ according to the specific grid standard.

5) Active power derating begins at 40°C when PF = ± 0.9 and MPPT $\geq$ V_{min}; at 45°C when PF = 1 and MPPT $\geq$ V_{min}; and at 50°C when PF = 1 and MPPT $\geq$ 700 Vdc.

6) Shade cover accessory required for installation angles of 75 degrees or less.

7) RSD wire-box only includes fuses and fuse holders on the positive polarity.

8) Firmware version 17.0 or later required.